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# Engaging and caring spaces for teachers and children

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*Publication date:*
2024

Link to publication

*Citation for pulished version (APA):*
Bøjer, B., & Rasmussen, L. R. (2024). *Engaging and caring spaces for teachers and children: Mediated intertwinements of pedagogy, physical space and IEQ*. Paper presented at ECER 2024 - Education in an age of uncertainty: memory and hope for the future. , Nicosia, Cyprus.

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# Engaging and caring spaces for teachers and children

## Mediated intertwinements of pedagogy, physical space and IEQ

Presentation at ECER 2024 by Bodil Hovaldt Bøjer & Lisa Rosén Rasmussen



# The issue

- How does the pedagogical use of the learning space impact IEQ?

- Can we, by enhancing knowledge about the relation between pedagogy, space and IEQ strengthen environmental competencies and thus create more engaging, inclusive, and caring space for both teachers and students



# Learning Space and IEQ

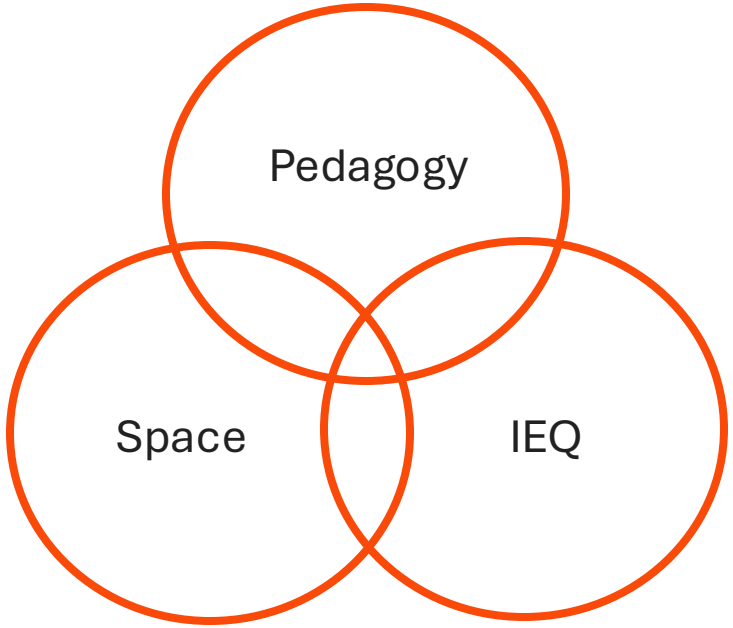
A Danish action research project about pedagogy, space, and IEQ and their interrelations from 2020-2024

Part 1: Literature review

Part 2: Action research:
Teacher training program

*Partners:*
*Aarhus Municipality*
*DCUM*
*Aarhus University*
*Royal Danish Academy*

*Supported by Realdania*

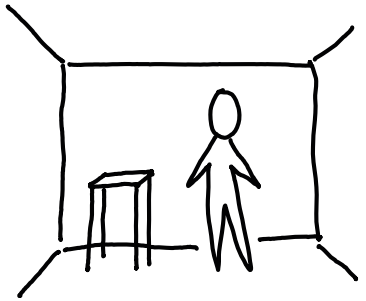




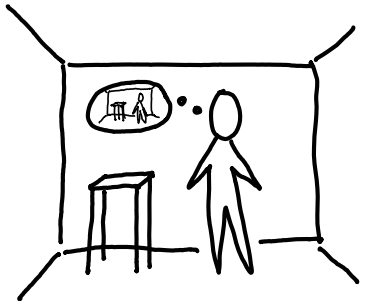

# Environmental competencies

Three ways to inhabit a space

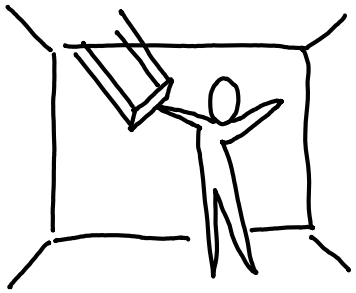
- **Being there**
  (space = neutral frame)



- **Being aware**
  (space has possibilities but I'm not sure how to use them actively)



- **Being competent**
  (I can redesign the space to match my practices – and my practices to match the space)



(Bøjer & Brøns, 2022)

# Theoretical approach

## New materialist readings of IEQ, space and pedagogy

In a new materialist perspective IEQ becomes part of a complex entanglement of the human and the non-human and matters of light, sound, and airflow may be seen as woven together with bodies and objects as part of what makes up the spatial and pedagogical practices of a learning environment (e.g. Charteris et al., 2017).

# Case study: Solbjerg Fritidsklub Computer Lab

## The challenge



A hybrid but pedagogical space

/ Fortnite, league of legends, Roblox

/ “It is our biggest educational workshop”

Pedagogically induced work on sound/noise

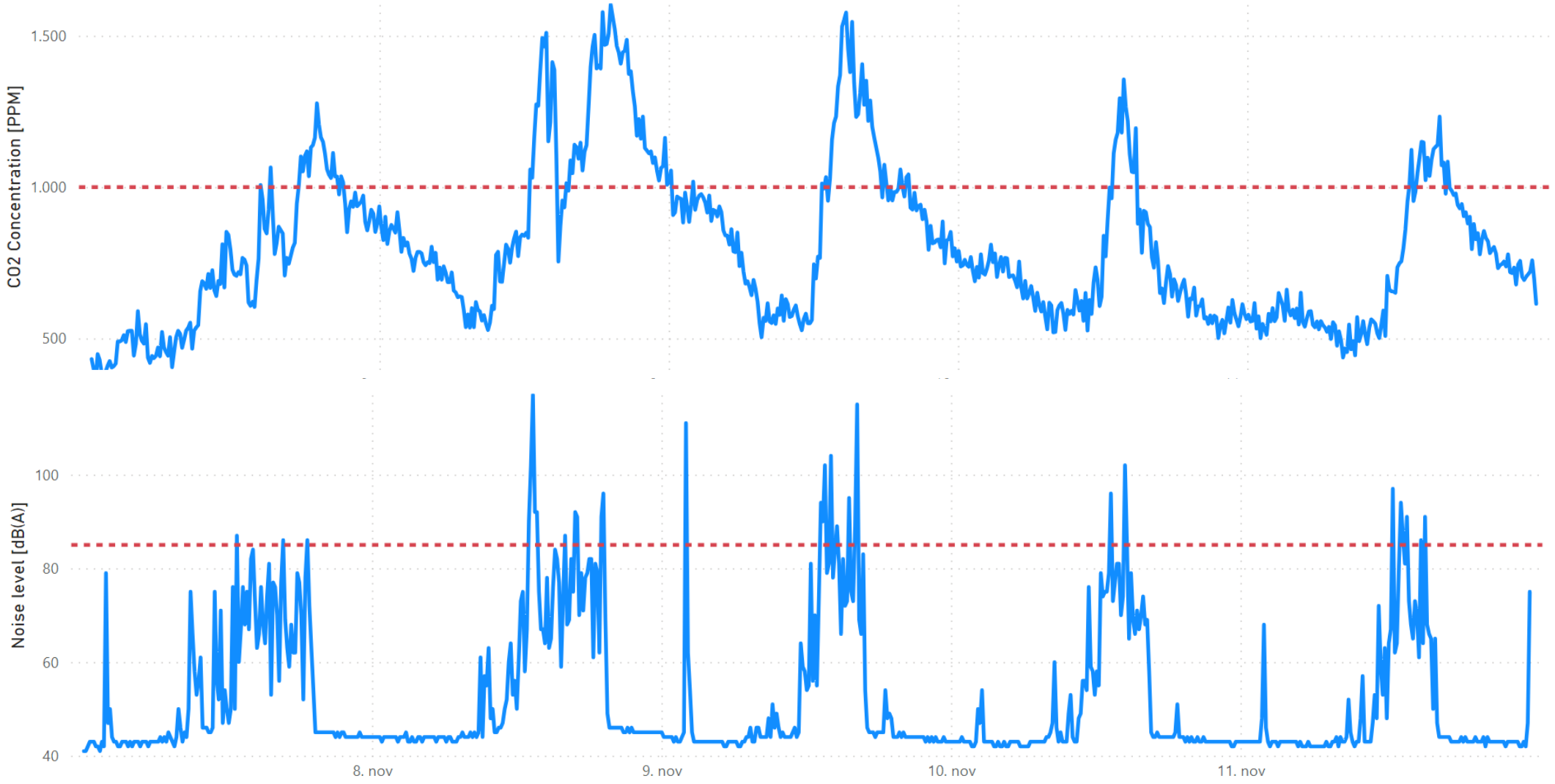
/ Headsets forbidden

/ Sound bats installed



# Case study: Solbjerg Fritidsklub Computer Lab

## Roomalyzers & CO2-particles [air flow and noise levels]






# Case study: Solbjerg Fritidsklub Computer Lab

Obligatory breaks [pedagogical re-organizations and IEQ]



# Case study: Solbjerg Fritidsklub Computer Lab

New children and a different indoor environment



# Case Study Solbjerg Fritidsklub: Findings

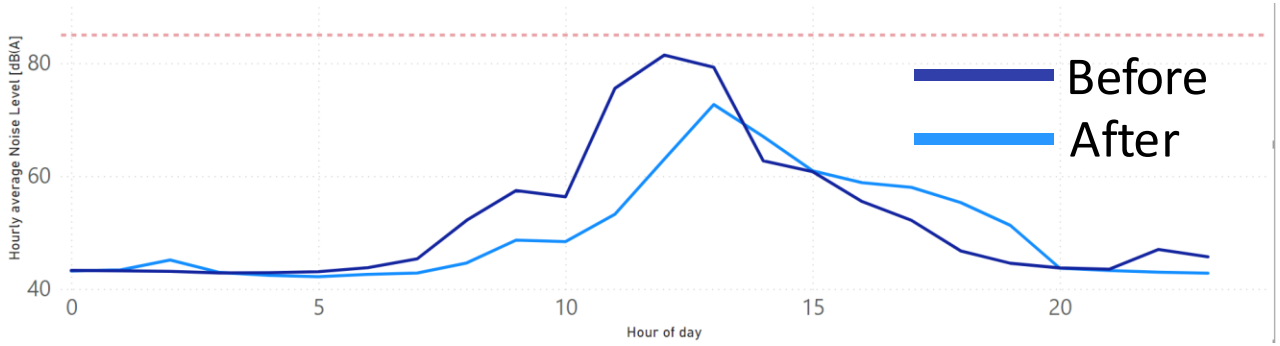
## Complex intra-actions and pedagogical transformations

Actions:

Acoustic panels, airing and pedagogical changes (e.g. controlled timeslots)

Results:

- Happier teachers and students
- Less noise and conflicts
- Change in clientele: more girls (btw. 1/4 and 1/3) and students with special needs



# **Caring spaces:**
# IEQ, inclusion and participation

New-materialist understandings of care and caring (Warren 2022):

*“Multiple, diverse components of early childhood settings continually produce care and caring relationships.”*

*“Bringing materials into view as producing and being produced alongside humans raises possibilities for considering how care matters and how matter cares in early childhood education environments.”*

# **Caring spaces:**
# IEQ, inclusion and participation

The Solbjerg case as an example of how:

- parameters of IEQ – acoustics and air quality – produce and are being produced alongside teaching strategies, teachers, students in ways which creates a caring and inclusive space giving room to different and also more vulnerable students.

- a teacher's sensitive approach to and knowledge of the supposed connectivity of IEQ, space, and pedagogy makes her able to transform a space from being highly uncomfortable and excluding to being caring and inclusive.

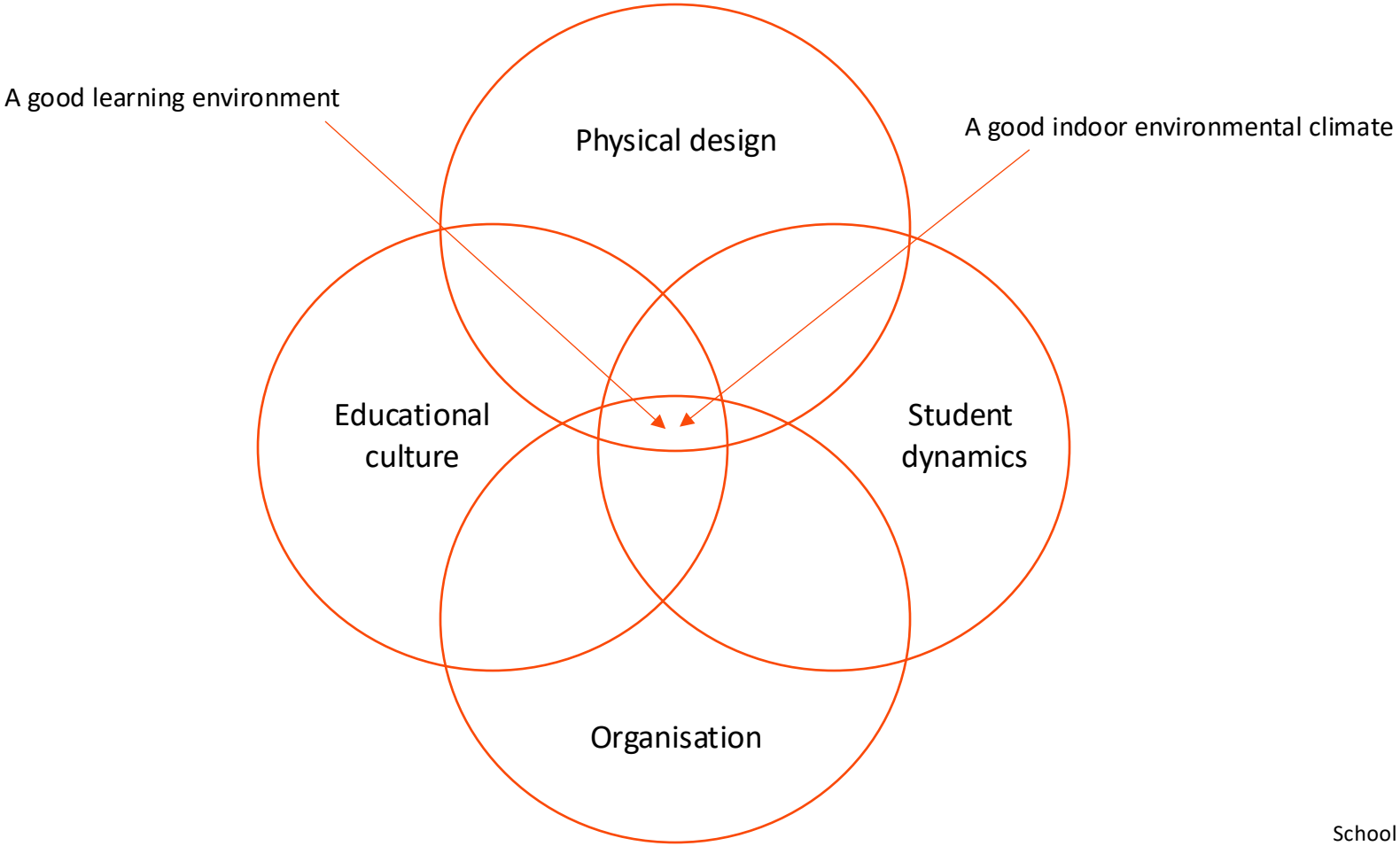
# Other examples

A common area and a reading space





# A good learning environment
## A good indoor environmental climate




School Climate Model by Neil Gislason, 2018
Adapted and translated by Bøjer & Rasmussen, 2024

# To think and do indoor environmental climate pedagogically

- Pedagogy, space, and indoor environmental climate as intervowen in the creation of a better, more caring and hence, democratic and including learning environment;
- Education in indoor enviromental qualities, space, and pedagogy as a way to empower the teachers;
- To balance these three parameters is a cross-disciplinary assignment and;
- Thinking and doing indoor environmental climate pedagogically is NOT a substitute for technical installations but a supplement!



# Thank you for listening!

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